

01 / EXPLORE SPATIAL THINKING

Nonograms

Shade squares to reveal a picture. Each clue gives the lengths of filled blocks in that row or column, in order. Separate blocks with at least one empty square. Mark known empty squares with a small cross.

EASY Reveal a five-by-five picture.

	2	4	4	4	2
1 1					
5					
5					
3					
1					

READING THE CLUES

A clue of 1 1 means one filled square, a gap of at least one square, then another filled square.

MEDIUM Reveal a seven-by-seven picture.

	1	2	3				
	1	2	2	7	3	3	1
1							
2							
3							
4							
3							
7							
5							

READING THE CLUES

Row clues are read left to right. Column clues are read top to bottom. The picture will emerge as your deductions meet.

Answers on page 4

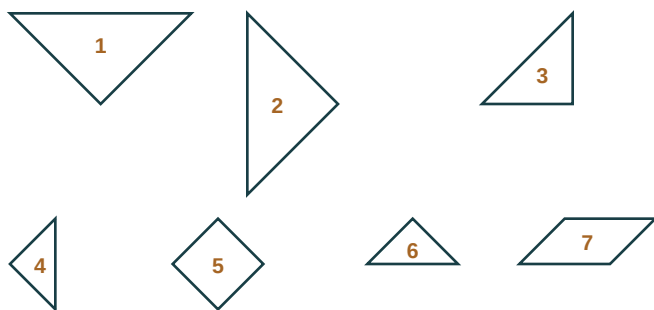
02 / EXPLORE SPATIAL THINKING

Tangrams

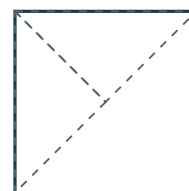
Draw dividing lines to fit all seven pieces into each outline. Use each piece once, without gaps or overlaps; you may turn or flip pieces. Each piece bank is at the same scale as its target. Dashed lines are starting hints.

EASY Make a square. Two large triangles are outlined.

YOUR SEVEN PIECES



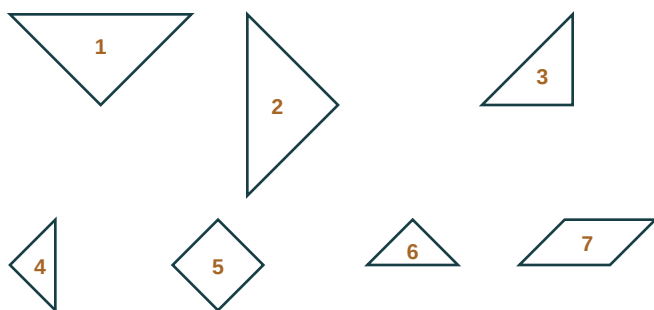
TARGET OUTLINE



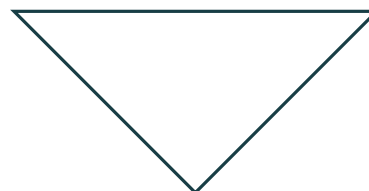
Sketch lightly; the solution shows one valid arrangement.

MEDIUM Make a large triangle, with no starting lines.

YOUR SEVEN PIECES



TARGET OUTLINE



Sketch lightly; the solution shows one valid arrangement.

Answers on page 4

03 / EXPLORE SPATIAL THINKING

Shape rotations

Imagine turning the target shape flat on the page, without flipping it over. Circle the option that matches the stated turn. All white and shaded squares must end up in the correct positions.

EASY Turn the target 90 degrees clockwise (a quarter-turn).

TARGET



A



B



C



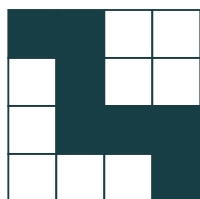
D



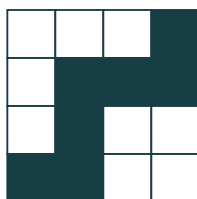
My choice: _____

MEDIUM Turn the target 270 degrees clockwise (three quarter-turns).

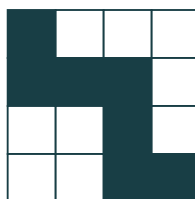
TARGET



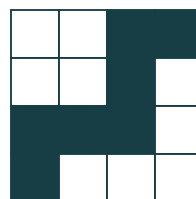
A



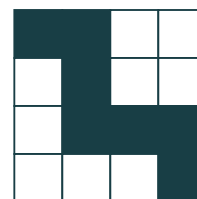
B



C



D



My choice: _____

Answers on page 4

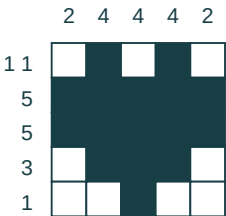
04 / YOUR ANSWERS

Answers & discoveries

Check your solutions and the explanations below. For visual arrangements and open-ended riddles, other answers count if they satisfy every rule.

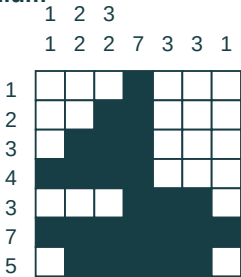
01 NONOGRAMS

Easy



A heart.

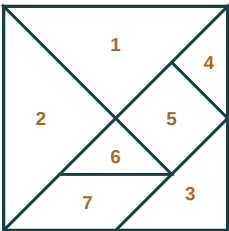
Medium



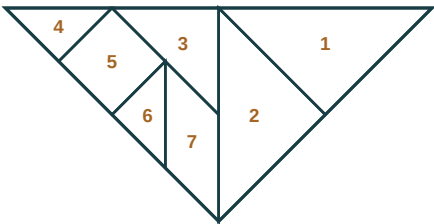
A sailboat.

02 TANGRAMS

Easy



Medium



03 SHAPE ROTATIONS

Easy

B - a quarter-turn clockwise.



Medium

C - three quarter-turns clockwise.



MY FAVOURITE TODAY

☐ Nonograms

☐ Tangrams

☐ Rotations